

AI FitTrack

7. MVC PATTERN

7.1 Introduction

AI FitTrack follows the MVC (Model–View–Controller) design pattern to organize the application into three separate components: Model, View, and Controller. MVC helps separate the user interface from the application logic and database operations. This makes the system easier to develop, maintain, debug, and modify.

7.2 MVC Components

1. Model

The Model is responsible for handling data and database operations.

Responsibilities:

- Connect with MySQL database.
- Store and retrieve user information.
- Manage BMI records.
- Manage workout and diet data.
- Manage calorie and progress records.

Example:

```
UserModel  
BMIModel  
WorkoutModel  
DietModel  
ProgressModel
```

2. View

The View represents the user interface of the AI FitTrack application.

Technologies Used: HTML, CSS, JavaScript, Bootstrap

Responsibilities:

- Display login and registration pages.
- Display user dashboard.
- Display BMI results.
- Display workout plans.
- Display diet plans.
- Display fitness progress.

3. Controller

The Controller acts as an intermediate layer between the Model and View. It receives requests from the user, processes them, communicates with the Model, and sends the result to the View.

Responsibilities:

- Process user requests.
- Validate input.
- Call appropriate Model functions.
- Perform application logic.

- Return results to the View.

Example:

```
UserController
BMIController
WorkoutController
DietController
ProgressController
```

7.3 MVC Architecture

```
USER
↓
VIEW – HTML / CSS / JavaScript
↓
CONTROLLER – PHP
↓
MODEL – PHP + MySQL
↓
DATABASE – MySQL
```

7.4 MVC Data Flow

The user interacts with the View through the web browser. The View sends the request to the Controller. The Controller processes the request and communicates with the Model. The Model performs database operations and returns the required data to the Controller. Finally, the Controller sends the result back to the View.

```
User → View → Controller → Model → MySQL Database → Model → Controller → View →
User
```

7.5 Advantages of MVC

- Separates application components clearly.
- Makes the code easier to maintain.
- Reduces code duplication.
- Makes debugging easier.
- Allows frontend and backend components to be modified independently.
- Supports future expansion of the AI FitTrack system.
- Improves code organization and reusability.

7.6 MVC in AI FitTrack

In AI FitTrack, the View displays the fitness interface, the Controller handles user requests and application logic, and the Model manages fitness data stored in MySQL.

Example: BMI Calculation

```
User enters Height & Weight
↓
View
↓
BMI Controller
↓
BMI Model
↓
Calculation
↓
```

Result



View

Thus, the MVC pattern provides a structured and maintainable architecture for the AI FitTrack application.